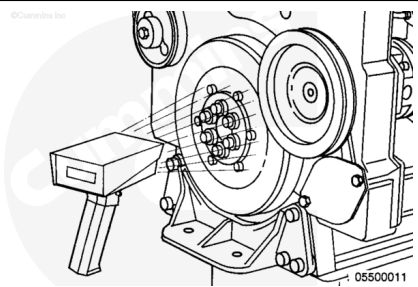


Trainings ▾

Pressure Test

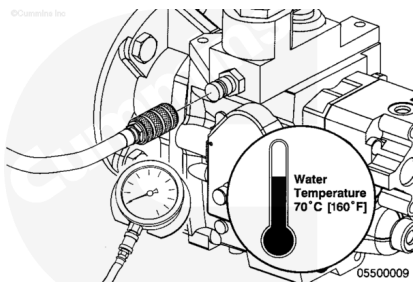
Use INSITE™ electronic service tool, the vehicle tachometer, or an optical tachometer, Part Number 3377462, to check the fuel pump pressure at a specified rpm. See the fuel pressure specifications in this section.



Use a pressure gauge and hose assembly capable of 2800 kPa [400 psi] with quick disconnect fitting, Part Number ST-435-7.

Crank the engine at 1500 rpm and measure the fuel pressure. The minimum fuel pressure **must** be 241 kPa [35 psi].

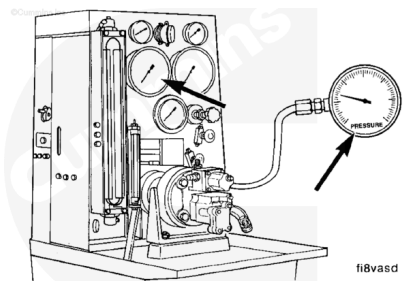
Start the engine and operate at high idle speed. Use the table below for specifications.



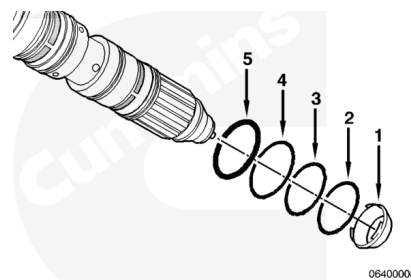
Minimum Fuel Pressure		
Engine High Idle Speed	kPa	psi
1500 rpm	1434	208
1800 rpm	1779	258
2100 rpm	2000	290

The fuel pressure will **not** change significantly with engine load.

If the fuel pump pressure is **not** within specifications, check the fuel pump on a fuel pump test stand.



If the fuel pump pressure is within specifications when checked on a fuel pump test stand, check the injector o-rings for damage. Refer to Procedure 006-026 in Section 6 (/qs3/pubsys2/xml/en/procedures/89/89-006-026-tr.html).

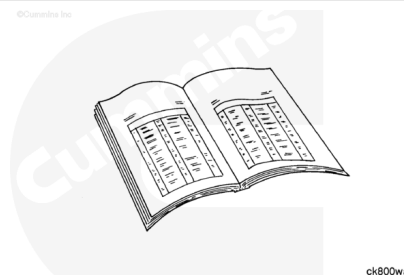


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery last.

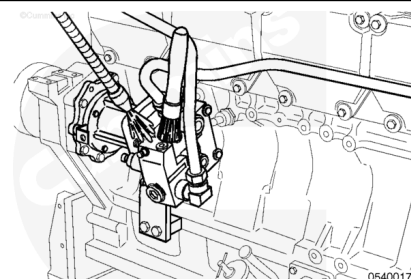
- Disconnect the battery cables. See equipment manufacturer service information.



Remove

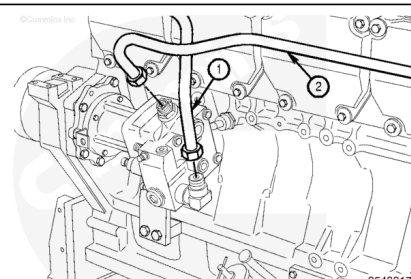
⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Clean the fuel pump and surrounding area.

Disconnect the fuel pump inlet hose (1). Disconnect the electronic fuel control valve supply hose (2).

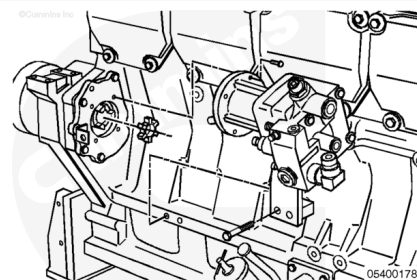


Remove the four mounting capscrews.

Remove the lower mounting bracket capscrews and remove the fuel pump.

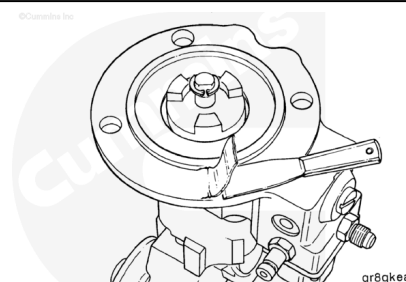
Remove the spider coupling.

Remove and discard the gasket.



Clean and Inspect for Reuse

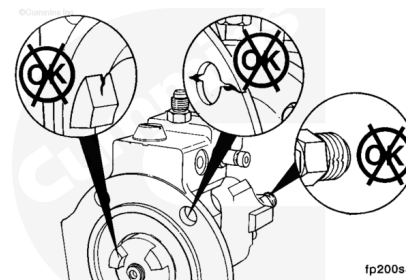
Clean the fuel pump mounting surface.



Inspect the fuel pump body and front support for cracks and other damage.

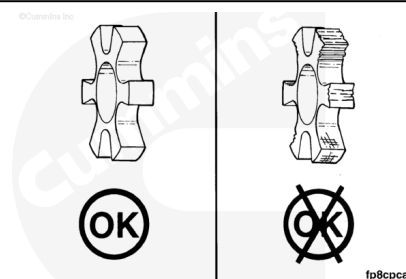
Inspect the fuel pump assembly for damaged or loose capscrews and fittings.

Inspect the drive coupling lugs for excessive wear or damage.



Inspect the spider coupling for cracks or damage.

Replace if damage or excessive wear is noted.



Install

Install the fuel pump drive spider coupling, fuel pump gasket, and fuel pump.

Completely install the fuel pump mounting capscrews, but do **not** tighten them.

Install the fuel pump support bracket.

Tighten the capscrews.

Torque Value:

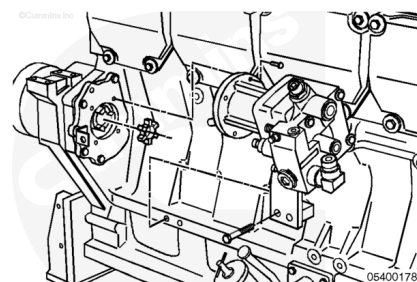
Bracket to Fuel Pump 100 n•m [75 ft-lb]

Torque Value:

Fuel Pump Mounting 70 n•m [52 ft-lb]

Torque Value:

Bracket to Block 70 n•m [52 ft-lb]



Install the fuel pump inlet hose (1) and the electronic fuel control valve supply hose (2).

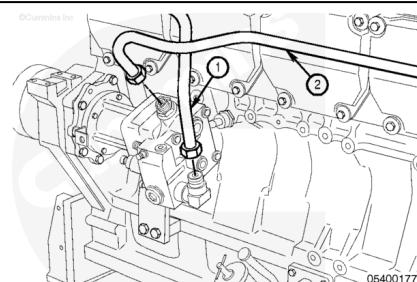
Tighten the hoses.

Torque Value:

Fuel Pump Inlet Hose 65 n•m [48 ft-lb]

Torque Value:

Fuel Control Supply Hose 90 n•m [66 ft-lb]



Prime

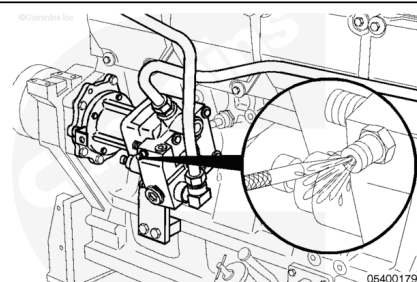
⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

Note : This procedure is normally **only** required when the fuel pump has been changed, the fuel filters are dry, or the fuel system has been drained.

If the fuel supply pump is dirty, clean the outside of the pump near the fuel inlet and outlet.

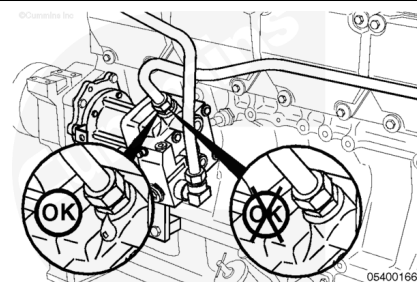
To reduce engine cranking time, depress the quick-disconnect fitting at the fuel pump outlet while cranking the engine until fuel is present.



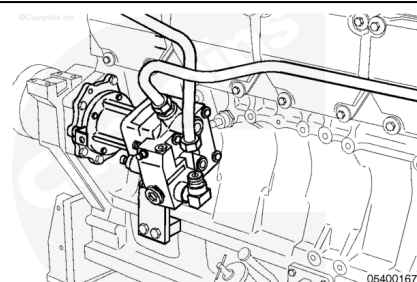
While the engine is cranking, loosen the electronic fuel control valve supply line at the top of the fuel pump. If fuel does **not** come out of the connection, the pump **must** be primed.

Tighten the fuel supply line.

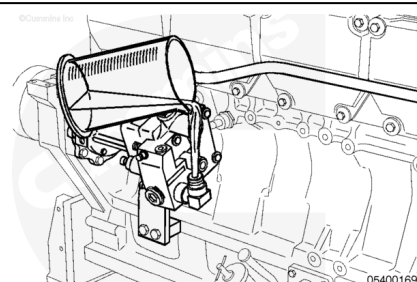
Torque Value: 90 n•m [66 ft-lb]



Remove the fuel pump inlet hose.



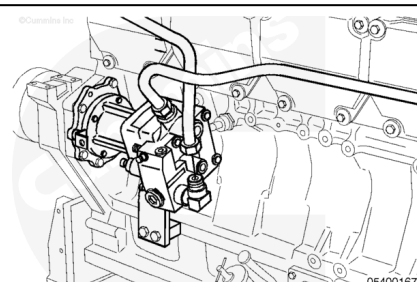
Use clean fuel and fill the fuel pump.



Install the fuel pump inlet hose to the fuel pump.

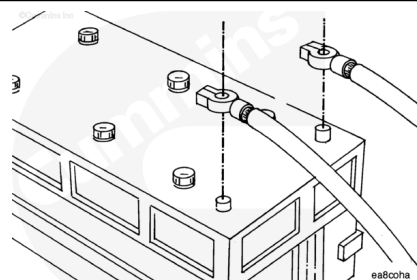
Tighten fuel supply hose.

Torque Value: 90 n•m [66 ft-lb]



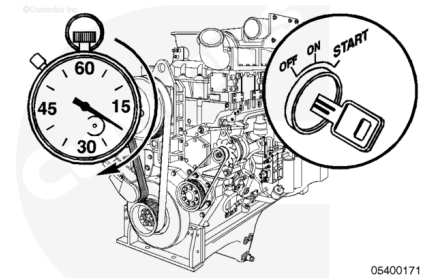
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery last.



- Connect the battery cables. See equipment manufacturer service information.

Crank the engine for 20 seconds. If the engine does **not** start within 20 seconds, wait 2 minutes and repeat the cranking procedure until the engine starts. The engine may run rough until the air is out of the system.



Finishing Steps

Start and operate the engine to normal operating temperature and check for leaks.

